SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

2

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

83

Code of Conduct:

I Pranav (192525420) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

	Problem identification 20%	Debugging 25%	Optimization 20%	Analysis 20%	Code Quality 15%	
Q1	1.5	2.5	1.5	2	1.5	9
Q2	1.5	2	1	2	1	7.5
Q3	1.5	2.5	2	1.5	~~2~~ 1.5	9
Q4	1	1.5	1.5	2	1.5	7.5
Q5	1.5	2.5	2	1.5	1	8.5

41.5/50

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Answer:

a) Subnet Identification

Subnet mask /26 (255.255.255.192) gives a block size of 256 - 192 = 64. This divides 192.168.10.0/24 into four /26 blocks: .0-.63, .64-.127, .128-.191 and .192-.255. The configured address 192.168.10.65 falls inside the second block, so it actually belongs to subnet 192.168.10.64/26 - not to 192.168.10.0/26 as required.

b) Network Address, Broadcast Address and Host Range

- For 192.168.10.64/26 -> Network address: 192.168.10.64, Broadcast address: 192.168.10.127, Valid host range: 192.168.10.65 - 192.168.10.126.
- For the required subnet 192.168.10.0/26 -> Network address: 192.168.10.0, Broadcast address: 192.168.10.63, Valid host range: 192.168.10.1 - 192.168.10.62.

c) Validity Check

The host IP 192.168.10.65 is a technically valid /26 host address, but it is valid only for the 192.168.10.64/26 subnet, not for 192.168.10.0/26. The gateway 192.168.10.1 is valid only for subnet 192.168.10.0/26. Because the host and the gateway sit in two different /26 subnets, the workstation cannot reach the gateway at Layer 3, so all off-subnet communication fails.

d) Configuration Errors Identified

- Host address (192.168.10.65) and default gateway (192.168.10.1) belong to two different /26 subnets - a classic subnet-boundary mismatch.
- The workstation was never actually placed inside the intended 192.168.10.0/26 network, even though the subnet mask was written correctly.

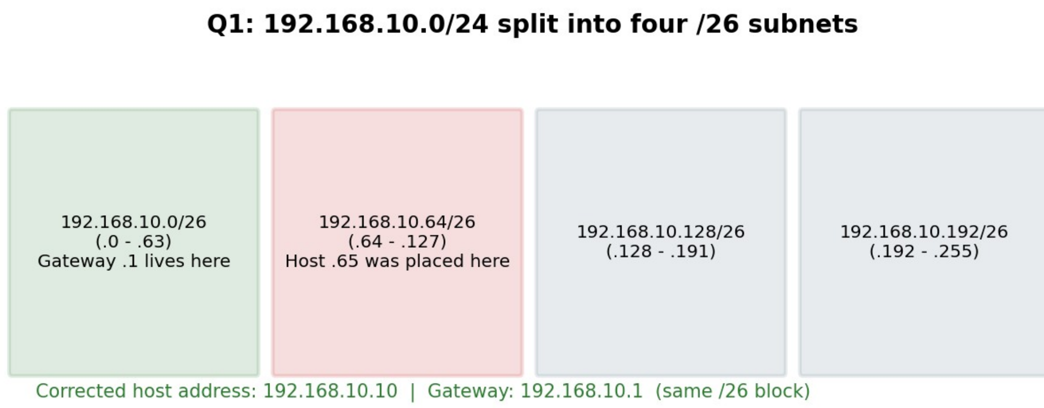
e) Optimized Configuration

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

All three values now fall inside the same subnet block (192.168.10.0 - 192.168.10.63), restoring communication with the rest of the LAN.

f) Usable Host Addresses After Correction

Usable hosts = 2^(32-26) - 2 = 2^6 - 2 = 62 usable addresses (192.168.10.1 to 192.168.10.62) in the 192.168.10.0/26 subnet.



2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

a) Full (Expanded) Notation

- Device A: 2001:db8::ff00:42:8329/64 -> 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1/64 -> 2001:0db8:0000:0000:0001:0000:0000:0001

b) Network Prefix of Each Device

With a /64 mask, the network prefix is the first 64 bits (first four hextets). Device A prefix = 2001:0db8:0000:0000::/64. Device B prefix = 2001:0db8:0000:0000::/64.

c) Same-Subnet Verification

Both devices resolve to the identical /64 prefix 2001:db8::/64, so - once correctly expanded - they DO belong to the same subnet. The addressing itself is therefore not the true fault; the earlier compressed notation only made the addresses look different.

d) Addressing / Prefix Mismatch Identified

The real issue is a documentation/interpretation error: the compressed forms were misread as belonging to different networks. Where a genuine mismatch does occur in practice, it is usually because one interface was mistakenly configured with a different prefix length (e.g. /127 or /48) instead of /64, which would place it in a different subnet even with a matching prefix pattern.

e) Optimized IPv6 Configuration

- Device A: 2001:db8::1/64
- Device B: 2001:db8::2/64

Both interfaces are explicitly set to unique host IDs on the single confirmed prefix 2001:db8::/64, removing any ambiguity.

f) Available Host Addresses in the Corrected Subnet

A /64 prefix leaves 64 bits for the interface ID: 2^64 = 18,446,744,073,709,551,616 possible addresses - practically unlimited for any LAN segment.

Q2: IPv6 address expansion and prefix comparison

Device A (compressed):	`2001:db8::ff00:42:8329/64`
Device A (expanded):	`2001:0db8:0000:0000:0000:ff00:0042:8329`
Device B (compressed):	`2001:db8:0:0:1::1/64`
Device B (expanded):	`2001:0db8:0000:0000:0001:0000:0000:0001`

Shared /64 network prefix: 2001:0db8:0000:0000::/64 -> same subnet

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

<u>Task</u>

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer:

a) Network, Broadcast and Host Range per Subnet

- /26 (mask 255.255.255.192, block 64): Network 192.168.1.0, Broadcast 192.168.1.63, Host range .1-.62 (62 usable).
- /27 (mask 255.255.255.224, block 32): Network 192.168.1.64, Broadcast 192.168.1.95, Host range .65-.94 (30 usable).
- /28 (mask 255.255.255.240, block 16): Network 192.168.1.96, Broadcast 192.168.1.111, Host range .97-.110 (14 usable).

b) Usable vs Unused Addresses (original plan)

Assume actual departmental needs are: Dept A = 55 hosts, Dept B = 10 hosts, Dept C = 28 hosts, Dept D = 20 hosts. Under the original fixed allocation, Dept B was given a /26 (62 usable) for only 10 hosts, wasting 52 addresses, while Dept D was given a /28 (14 usable) for 20 hosts, leaving it 6 addresses short.

c) Inefficient Allocations Identified

- Over-allocation: a large /26 block assigned to a small department (Dept B) - address waste.
- Under-allocation: a small /28 block assigned to a department needing more hosts (Dept D) - address shortage.

d) Debugging the Addressing Plan

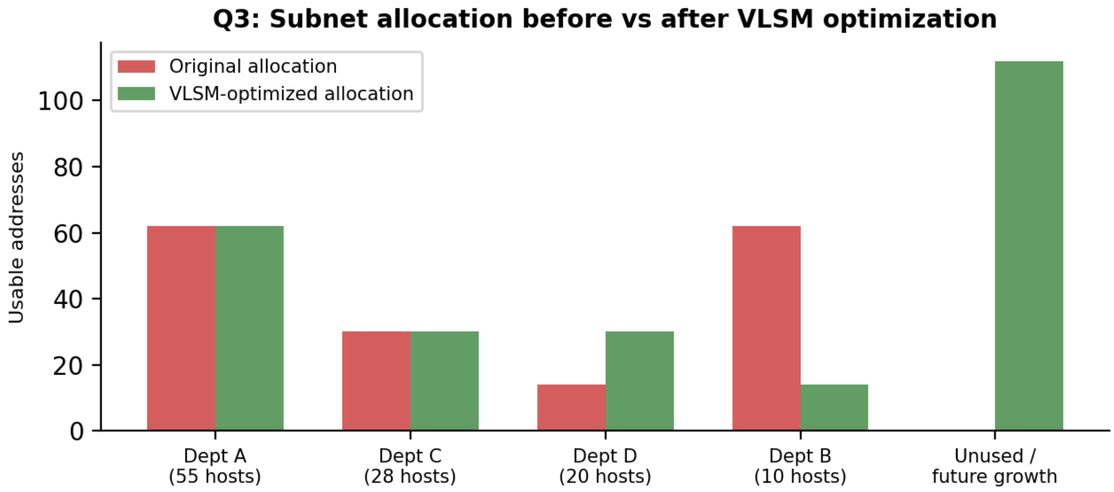
The plan mistakenly matched subnet size to department "type" rather than to the actual host count, causing the reported symptom: some departments run out of addresses while others sit on large unused ranges.

e) Recommended VLSM Redesign

Sort departments by host requirement (largest first) and assign the smallest subnet that satisfies each need:

- Dept A (55 hosts) -> 192.168.1.0/26 (62 usable)
- Dept C (28 hosts) -> 192.168.1.64/27 (30 usable)
- Dept D (20 hosts) -> 192.168.1.96/27 (30 usable)
- Dept B (10 hosts) -> 192.168.1.128/28 (14 usable)

This uses only 136 addresses precisely where needed and leaves 192.168.1.144 - 192.168.1.255 (112 addresses) free for future growth, eliminating both the shortage and the waste.



4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer:

a) Full (Expanded) Notation

- Device A: 2001:db8::ff00:42:8329/64 -> 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1/64 -> 2001:0db8:0000:0000:0001:0000:0000:0001

b) Network Prefix of Each Device

Taking the first 64 bits of each expanded address, both Device A and Device B share the same network prefix: 2001:0db8:0000:0000::/64.

c) Same-Subnet Verification

Since the prefixes match exactly, the two devices are logically on the same /64 subnet, confirming that basic IPv6 addressing is not the source of the failure.

d) Addressing / Prefix Mismatch Identified

No genuine prefix mismatch exists once the addresses are expanded. In real deployments this failure pattern typically comes from one host being manually assigned a non-standard prefix length instead of /64, or from a stale/duplicate interface ID left over from a previous configuration.

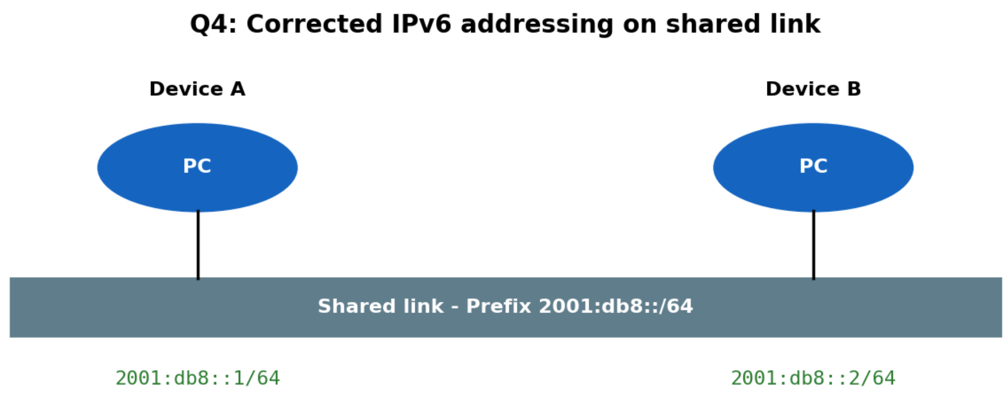
e) Optimized IPv6 Configuration

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Assigning clearly distinct, easy-to-read host IDs on the confirmed 2001:db8::/64 prefix removes any risk of address duplication or misreading.

f) Available Host Addresses in the Corrected Subnet

With a /64 prefix, 2^64 = 18,446,744,073,709,551,616 host addresses are available on the link.



5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

<u>Task</u>

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer:

a) Matching Entry via Longest-Prefix Match

Destination 192.168.2.100 is checked against every entry: it does not fall inside 192.168.1.0/24, it does fall inside 192.168.2.0/24, and the default route 0.0.0.0/0 always matches everything. Between the two real matches, the router picks the one with the longest prefix - here 192.168.2.0/24 (24 bits) beats the default route (0 bits).

b) Next-Hop Selected

The selected entry, 192.168.2.0/24, forwards packets to next-hop 10.0.0.3.

c) Destination Subnet Verification

192.168.2.100 lies inside the valid host range of 192.168.2.0/24 (192.168.2.1 - 192.168.2.254), confirming this is the intended matching subnet.

d) Routing Configuration Errors Identified

The routing table entries themselves are logically correct. If packets still fail to arrive, the fault is most likely outside the table: next-hop 10.0.0.3 unreachable or its interface down, a missing return route back to the source, or an access-list silently dropping the traffic. A common configuration mistake that produces exactly this symptom is the 192.168.2.0 entry being mistyped with a longer mask (e.g. /25), which would push .100 out of range and cause it to fall through to the default route instead.

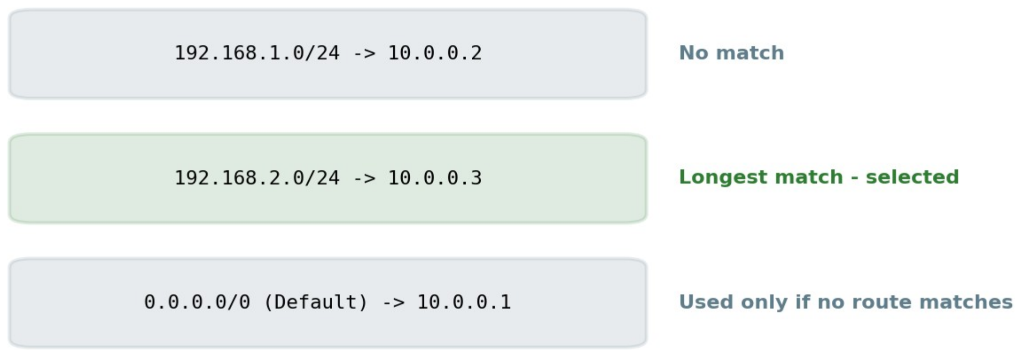
e) Routing Table Optimization

- Confirm the entry reads exactly 192.168.2.0/24 -> 10.0.0.3.
- Verify next-hop reachability with ping/traceroute and check the link/interface state.
- Add a floating static or dynamic backup route for redundancy so a single link failure does not blackhole traffic.

f) Route Used When No Entry Matches

If no specific destination entry matches, the router falls back to the Default Route, forwarding the packet to next-hop 10.0.0.1.

Q5: Longest-prefix-match decision for 192.168.2.100



RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic	Uses appropriate	Limited debugging	Unable to debug or

		debugging techniques effectively to resolve issues(2.5)	debugging methods with minor errors(2)	approach; requires guidance(1.5)	resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification (1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation